LAPORAN PRAKTIKUM DATABASE SYSTEMS

CHAPTER 6: SQL DATA DEFINITION (EXERCISES 6.10 & 6.11)

Mata Kuliah: Basis Data / Database Systems
Buku Acuan: Database Systems: A Practical Approach to Design, Implementation, and Management (Thomas Connolly & Carolyn Begg)
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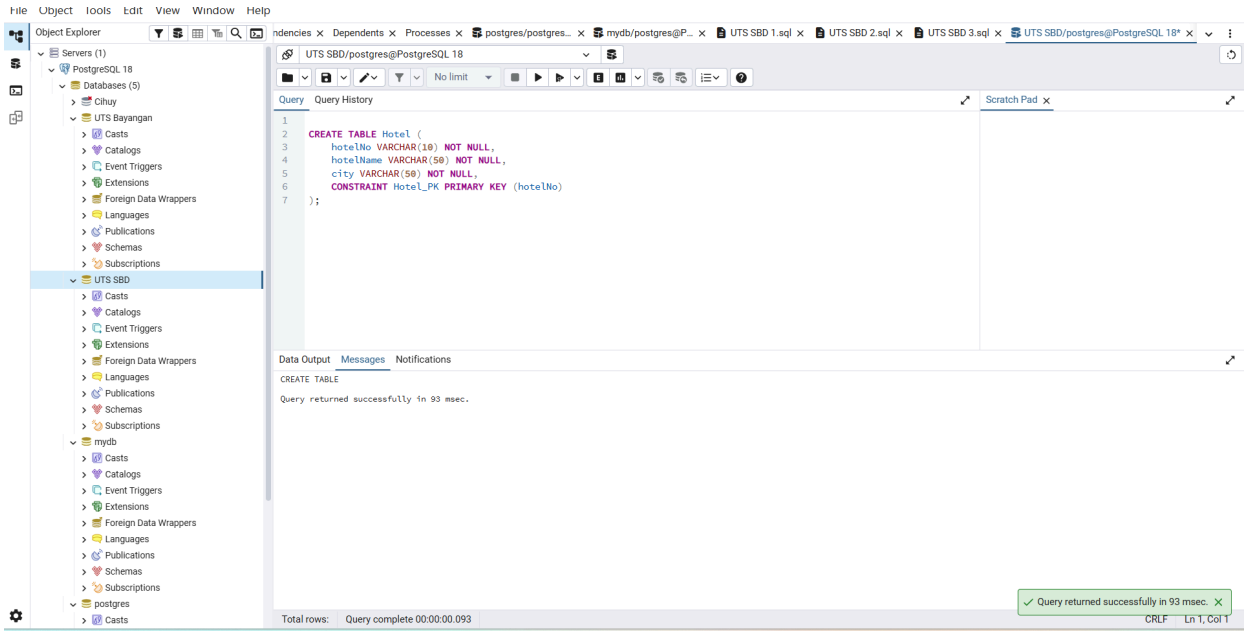
1. PENGERJAAN SOAL 6.10: PEMBUATAN TABEL `Hotel`

2.1 Soal 6.10

2.2 Penjelasan Fitur Integritas

- **`NOT NULL`**: Memastikan atribut `hotelNo`, `hotelName`, dan `city` wajib diisi dan tidak boleh bernilai NULL.
- **`PRIMARY KEY (hotelNo)`**: Memastikan *Entity Integrity*, yaitu setiap hotel memiliki identitas unik `hotelNo` dan tidak ada duplikasi.

2.3 Dokumentasi Eksekusi Query



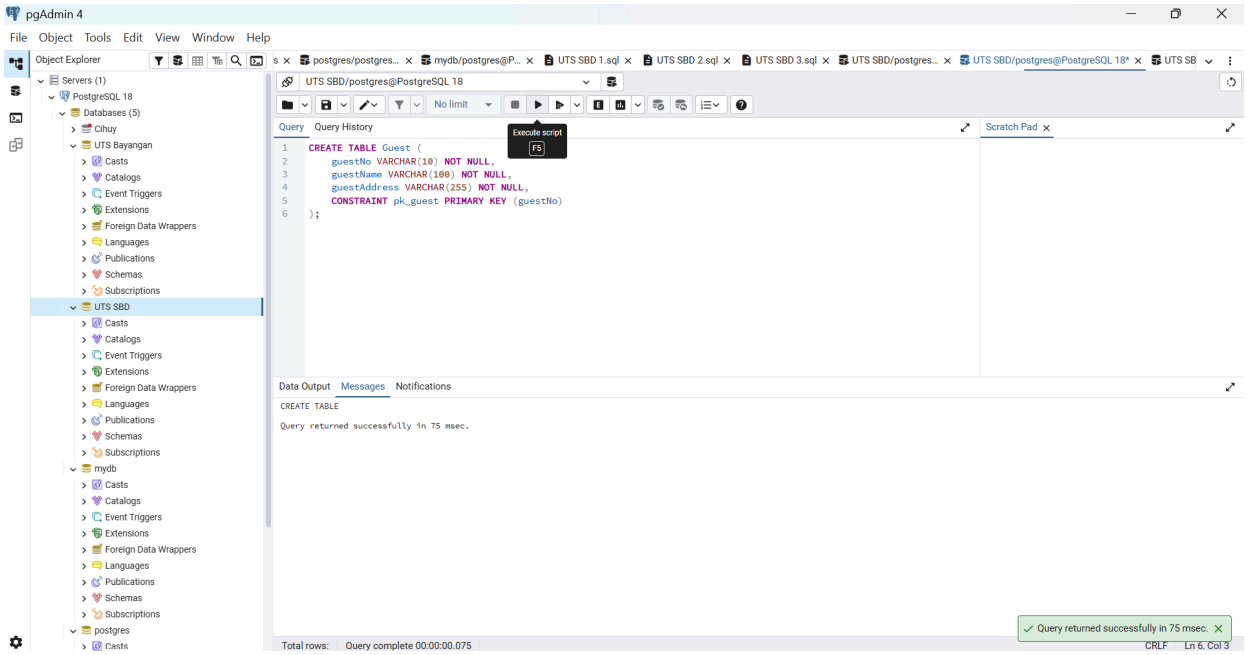
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3. PENGERJAAN SOAL 6.11: PEMBUATAN TABEL `Room`, `Guest`, DAN `Booking`

3.1 Soal 6.11

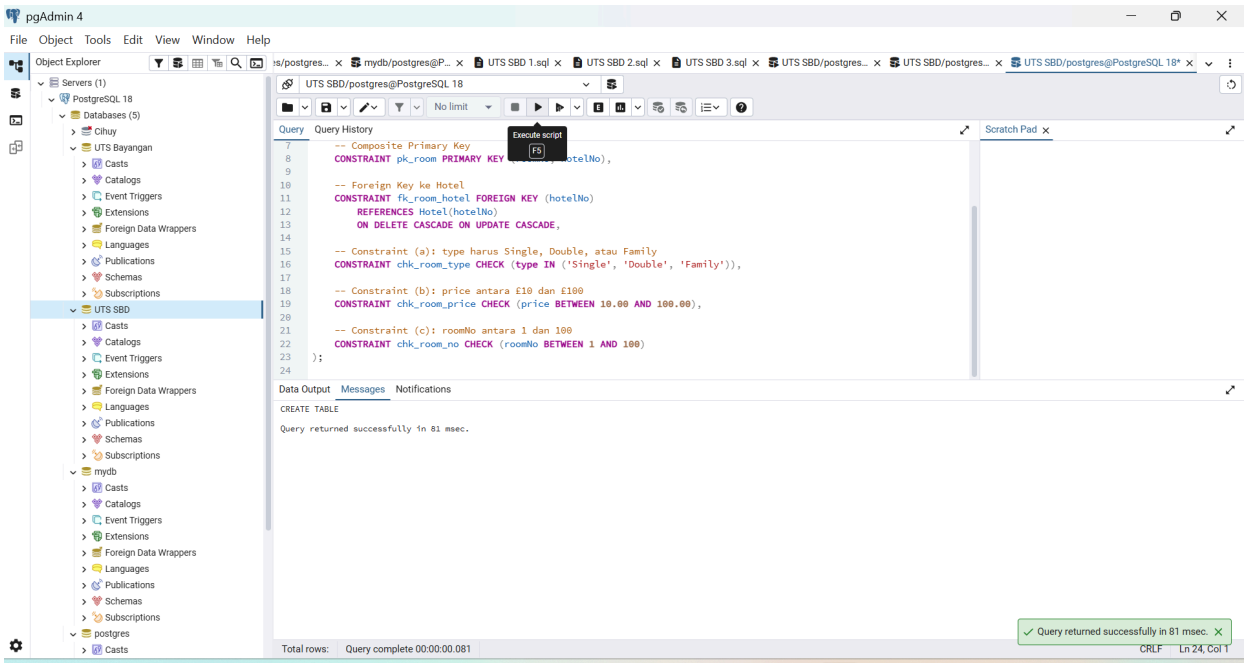
- (a) `type` must be one of Single, Double, or Family.
- (b) `price` must be between £10 and £100 (10 <= price <= 100).
- (c) `roomNo` must be between 1 and 100 (1 <= roomNo <= 100).
- (d) `dateFrom` and `dateTo` must be greater than today's date (dateFrom > CURRENT_DATE dan dateTo > CURRENT_DATE).
- (e) The same room cannot be double-booked.
- (f) The same guest cannot have overlapping bookings.

3.2 Pembuatan Tabel `Guest`



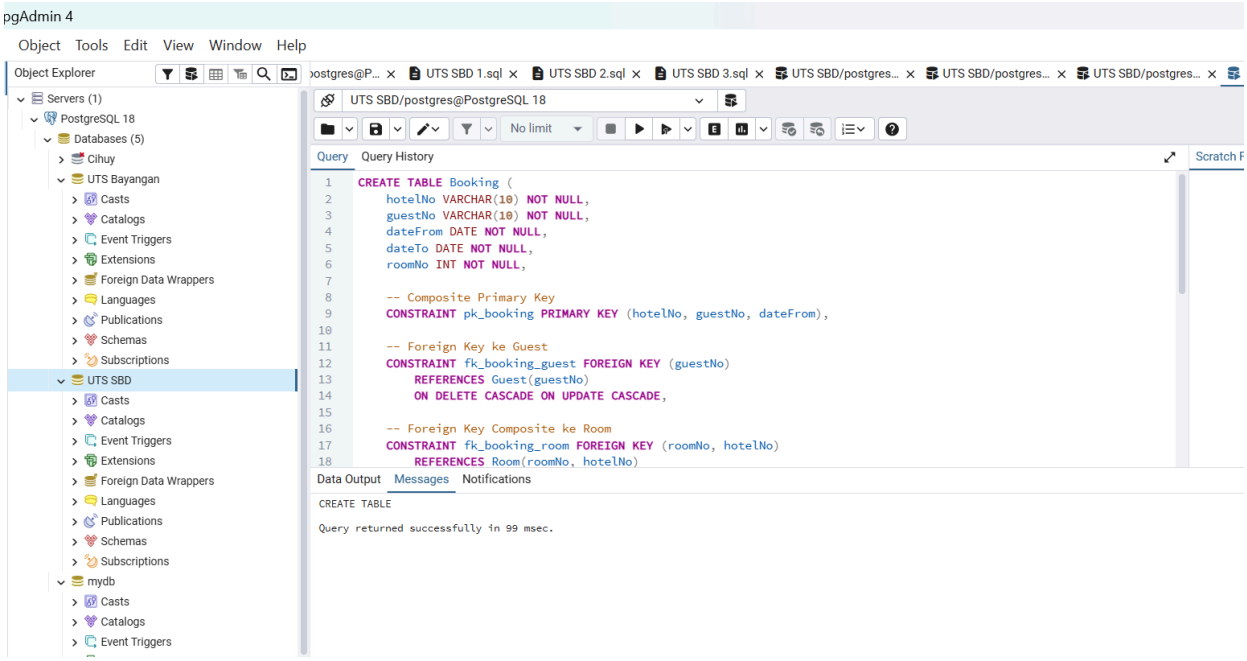
3.3 Pembuatan Tabel Room

Mencakup constraint **(a)**, **(b)**, dan **(c)**:



3.4 Pembuatan Tabel `Booking`

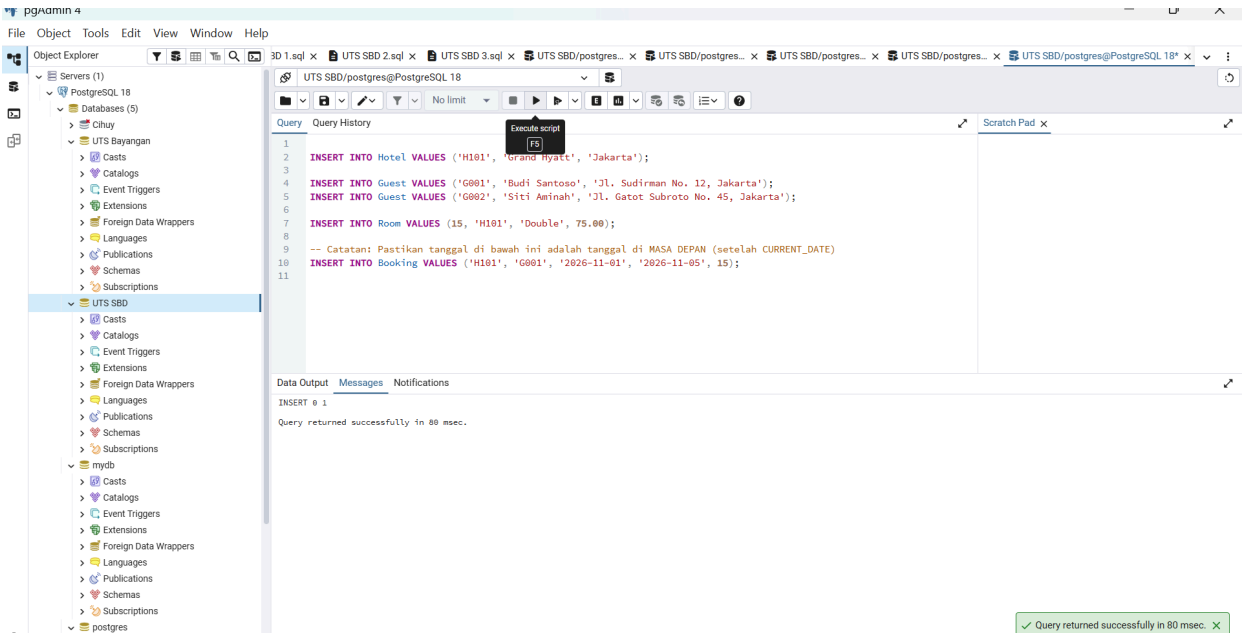
Mencakup constraint **(d)**, **(e)**, dan **(f)**.



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4. DOKUMENTASI PENGUJIEN CONSTRAINT (TEST CASES)

4.1 Uji Coba Input Data Valid



4.2 Uji Coba Constraint Violation (Skenario Error)

A. Pelanggaran Constraint (a) Tipe Kamar Tidak Valid

Query Query History

```
1  INSERT INTO Room VALUES (20, 'H101', 'Executive', 50.00);
```

Data Output Messages Notifications

```
ERROR:  new row for relation "room" violates check constraint "chk_room_type"
Failing row contains (20, H101, Executive, 50.00).

SQL state: 23514
Detail: Failing row contains (20, H101, Executive, 50.00).
```

B. Pelanggaran Constraint (b) Harga di Luar Jangkauan

Query | Query History

```
1  INSERT INTO Room VALUES (20, 'H101', 'Executive', 50.00);
2
3  INSERT INTO Room VALUES (21, 'H101', 'Single', 150.00);
4
5
6  INSERT INTO Room VALUES (105, 'H101', 'Family', 90.00);
7
8  INSERT INTO Booking VALUES ('H101', 'G001', '2020-01-01', '2020-01-05', 15);
9
10
11 INSERT INTO Booking VALUES ('H101', 'G002', '2026-11-03', '2026-11-07', 15);
```

Data Output | Messages | Notifications

```
ERROR:  new row for relation "room" violates check constraint "chk_room_price"
Failing row contains (21, H101, Single, 150.00).

SQL state: 23514
Detail: Failing row contains (21, H101, Single, 150.00).
```

C. Pelanggaran Constraint (c) Nomor Kamar di Luar Jangkauan

Query | Query History

```
1  INSERT INTO Room VALUES (20, 'H101', 'Executive', 50.00);
2
3  INSERT INTO Room VALUES (21, 'H101', 'Single', 150.00);
4
5
6  INSERT INTO Room VALUES (105, 'H101', 'Family', 90.00);
7
8  INSERT INTO Booking VALUES ('H101', 'G001', '2020-01-01', '2020-01-05', 15);
9
10
11 INSERT INTO Booking VALUES ('H101', 'G002', '2026-11-03', '2026-11-07', 15);
```

Data Output | Messages | Notifications

```
ERROR:  new row for relation "room" violates check constraint "chk_room_no"
Failing row contains (105, H101, Family, 90.00).

SQL state: 23514
Detail: Failing row contains (105, H101, Family, 90.00).
```

D. Pelanggaran Constraint (d) Tanggal Booking Lampau



Query | Query History

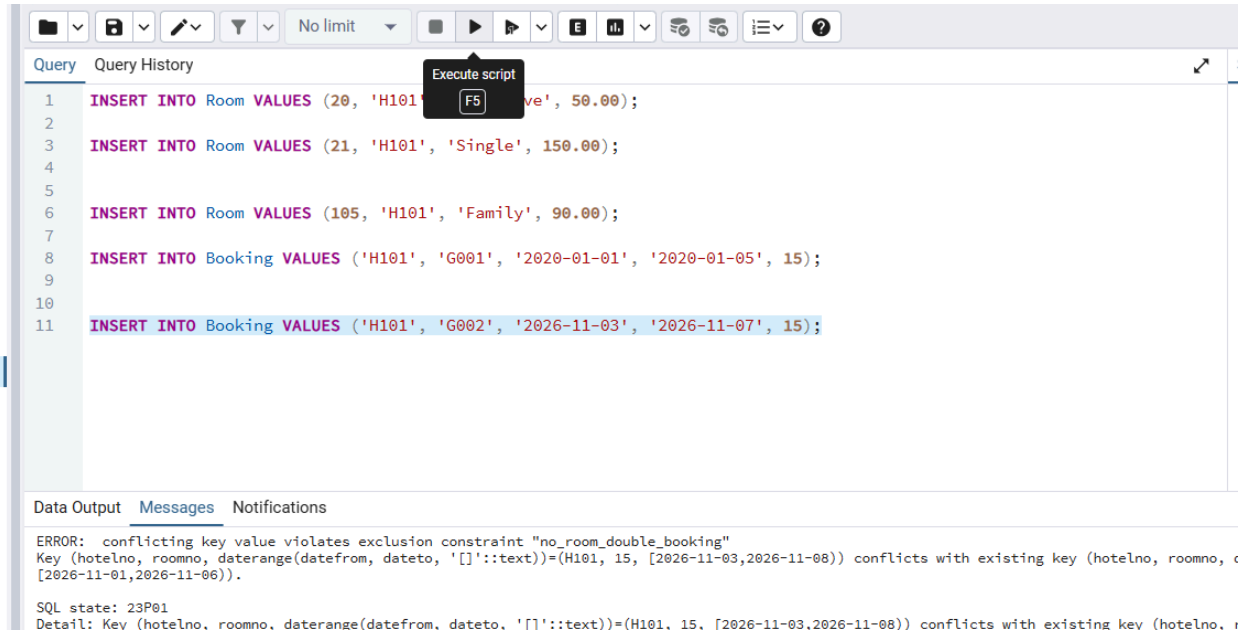
```
1  INSERT INTO Room VALUES (20, 'H101', 'Executive', 50.00);
2
3  INSERT INTO Room VALUES (21, 'H101', 'Single', 150.00);
4
5
6  INSERT INTO Room VALUES (105, 'H101', 'Family', 90.00);
7
8  INSERT INTO Booking VALUES ('H101', 'G001', '2020-01-01', '2020-01-05', 15);
9
10
11 INSERT INTO Booking VALUES ('H101', 'G002', '2026-11-03', '2026-11-07', 15);
```

Data Output | Messages | Notifications

```
ERROR:  new row for relation "booking" violates check constraint "chk_booking_dates"
Failing row contains (H101, G001, 2020-01-01, 2020-01-05, 15).

SQL state: 23514
Detail: Failing row contains (H101, G001, 2020-01-01, 2020-01-05, 15).
```

E. Pelanggaran Constraint (e) Double-Booking Kamar yang Sama



No limit | Execute script (F5)

Query | Query History

```
1  INSERT INTO Room VALUES (20, 'H101', 'Executive', 50.00);
2
3  INSERT INTO Room VALUES (21, 'H101', 'Single', 150.00);
4
5
6  INSERT INTO Room VALUES (105, 'H101', 'Family', 90.00);
7
8  INSERT INTO Booking VALUES ('H101', 'G001', '2020-01-01', '2020-01-05', 15);
9
10
11 INSERT INTO Booking VALUES ('H101', 'G002', '2026-11-03', '2026-11-07', 15);
```

Data Output | Messages | Notifications

```
ERROR:  conflicting key value violates exclusion constraint "no_room_double_booking"
Key (hotelno, roomno, daterange(datefrom, dateto, '[]'::text))=(H101, 15, [2026-11-03,2026-11-08)) conflicts with existing key (hotelno, roomno, d
[2026-11-01,2026-11-06)).

SQL state: 23P01
Detail: Key (hotelno, roomno, daterange(datefrom, dateto, '[]'::text))=(H101, 15, [2026-11-03,2026-11-08)) conflicts with existing key (hotelno, r
```